Architectural & Technology Stack Report

Project: Campaign OS (Namma Brew) **Date:** October 9, 2026

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1. Executive Summary

This document provides a comprehensive technical overview of the Campaign OS application. The system is built on a highly modern, bleeding-edge web stack optimized for performance, aesthetic rendering, and advanced AI workflow integration.

2. Core Framework & Engine

- **Next.js 16.4.0 (App Router):** Serves as the foundational framework, utilizing modern routing paradigms and seamlessly bridging server-side API routes with client-side interactivity.
- **React 19.3.0:** Leverages the latest React ecosystem, employing concurrent rendering and state-of-the-art hooks.
- **TypeScript (v5+):** Enforces strict typing across the entire codebase. Custom interfaces (e.g., `CampaignStrategy`, `LockedOfferFacts`) ensure deterministic data flows and eliminate runtime type errors.

3. Styling & Presentation Layer

- **Tailwind CSS v4 + Turbopack:** Utilizes the newest Tailwind compiler directly integrated with Turbopack for near-instantaneous compilation and highly granular styling control.
- **Typography:** Integrates curated font pairings via CSS variables:
  - **Playfair Display:** Utilized for elegant, high-impact semantic headings.
  - **Caveat:** Utilized for handwritten, cursive accents to enhance visual warmth.
- **Lucide React:** Powers a scalable, consistent SVG iconography system.
- **Glassmorphism & Micro-Interactions:** Heavily relies on translucent backdrops (`backdrop-blur`), deep shadows, and

custom CSS keyframes to create a premium, tactile user interface.

4. State Management & Workflow Architecture

- **Monolithic State Machine:** The entire 10-step campaign generation workflow is managed within `src/app/campaign/page.tsx` via a complex, centralized React State machine. It actively controls UI rendering logic and data propagation (briefs, facts, strategies) without requiring browser reloads.
- **Client-Side Event Strictness:** As a Single Page Application (SPA), all navigational components explicitly enforce `type="button"` and `e.preventDefault()` paradigms to prevent default form submittals that would otherwise disrupt the application state.

5. Backend AI Integration & Security

- **Agnes AI API Integration:** Uses Edge-compatible Next.js Route Handlers to securely communicate with the Agnes AI infrastructure (`apihub.agnes-ai.com`).
- **Self-Healing Validation Loop:** The backend architecture employs an innovative self-correcting loop. Prior to serving generated content to the frontend, an AI-driven compliance validator strictly audits the output against locked business facts. Any hallucinated claims (e.g., false discounts or locations) trigger an automatic, silent regeneration sequence.
- **Network Tunneling Exception:** Next.js 16 enforces strict host header validations. A targeted security exception (`allowedDevOrigins`) is configured in `next.config.ts` to authorize cross-origin requests routing through ngrok tunnels, preventing `ERR_NGROK_8012` crashes.

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